The Structure of Matter

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Introduction

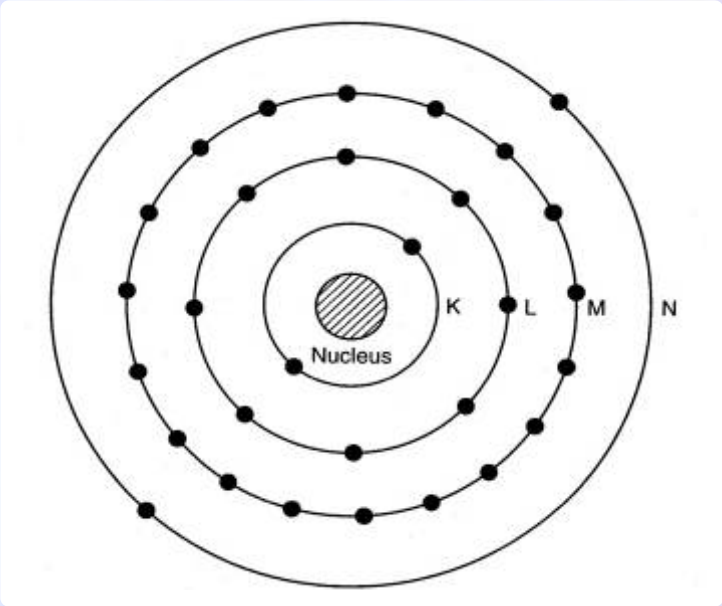
- Element - a substance that cannot be broken down by ordinary chemical processes into simpler substances
  - 92 naturally occurring elements
  - Another dozen or so elements have been produced artificially, best known is plutonium
- Compound - two or more elements chemically linked in definite proportions, e.g. water, H_2O
- Atom - matter consists of atoms, each element having its own characteristic atom

Structure of the Atom

- Every atom consists of protons, neutrons and electrons
- Proton – one unit of positive electrical charge and mass of approximately one atomic mass unit (1u)
- Electron - one unit of negative charge and mass of 1/1840 u
- Neutron - electrically neutral and mass of approximately 1 u

Model of the Atom

- Neutrons and protons form central nucleus
- Electrons rotate around nucleus in various orbits, normally referred to as shells
- K shell - 2 electrons
- L shell - 8 electrons
- M shell -18 electrons
- N shell - 32 electrons



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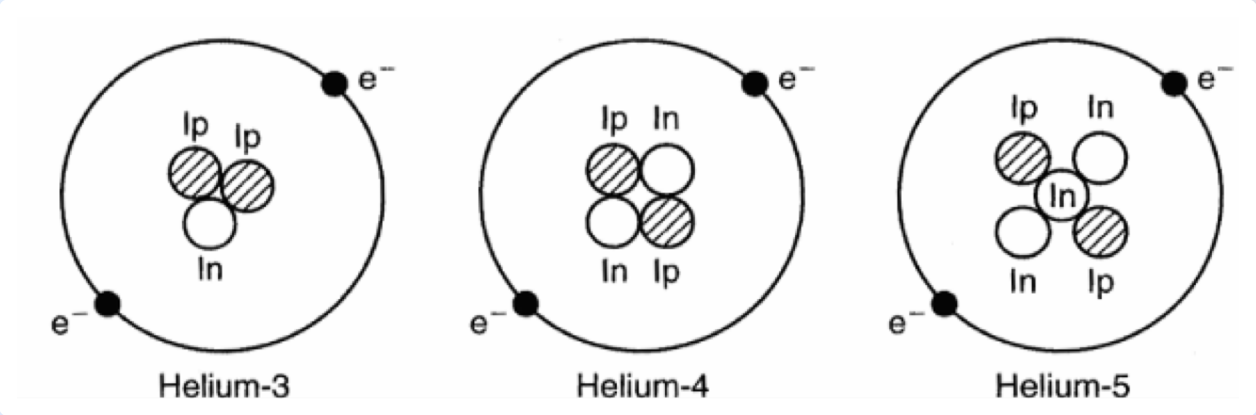
Isotopes and Mass Number

- All atoms of an element have the same number of protons but they can occur with different numbers of neutrons
- So an element can have several types of atoms – these are called isotopes and differ from each other in terms of their mass number (A)
- A = Z + number of neutrons

The Isotopes of Helium

- Helium can occur in three isotopic forms:



  - Helium - 3 (He-3) contains 2 protons and 1 neutron
  - Helium - 4 (He-4) contains 2 protons and 2 neutrons
  - Helium - 5 (He-5) contains 2 protons and 3 neutrons
- The term nuclide means any isotope of any element
- Note: All isotopes of an element are *chemically* identical

Radioactivity and radiation

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Atomic Instability

- Atoms of a few naturally-occurring elements are unstable - undergo spontaneous transformation into more stable atoms by a process known as radioactive decay
- Such substances are said to be radioactive
- In radioactive decay three main types of radiation are emitted: alpha, beta and gamma
- Natural radioactivity was first recognized by Becquerel in 1896. He gives his name to the SI unit of activity (see later lecture)

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Alpha, Beta, Gamma Radiations

- Alpha (α) radiation consists of helium nuclei (2 protons + 2 neutrons)
- α particle – mass 4 u, charge +2
- Beta (β) radiation consists of high-speed electrons originating in nucleus
- β particle - mass 1/1840 u, charge -1
- Gamma (γ) radiation results from changes in nucleus: consists of quanta of energy
- Gamma rays - electromagnetic radiation, energy inversely proportional to wavelength, $E \alpha 1/\lambda$

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Radioactive Decay Mechanisms

- α-emission in which number of protons and neutrons in nucleus are each reduced by 2
- β-emission - a neutron changes into a proton by emitting a high-speed electron(β particle):

$$^{1}_{0}n \longrightarrow ^{1}_{1}p + \beta^{-}$$

- Positron emission - proton in nucleus ejects positive electron(β^{+}) to become neutron
- Electron capture - an inner electron is captured by nucleus resulting in conversion of a proton into a neutron:

$$^{1}_{1}p + e^{-} \longrightarrow ^{1}_{0}n$$

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Radioactive Decay Law

- Radioactive decay - a random, statistical process governed by the mathematical law

$$N_t = N_0 e^{-\lambda t}$$

- The half-life ($T_{½}$) of a radioactive species is the time required for ½ of nuclei in sample to decay

$$T_{½} = 0.693/\lambda$$

- The disintegration rate (activity) is proportional to the number of unstable nuclei

$$A_t = A_0 e^{-\lambda t}$$

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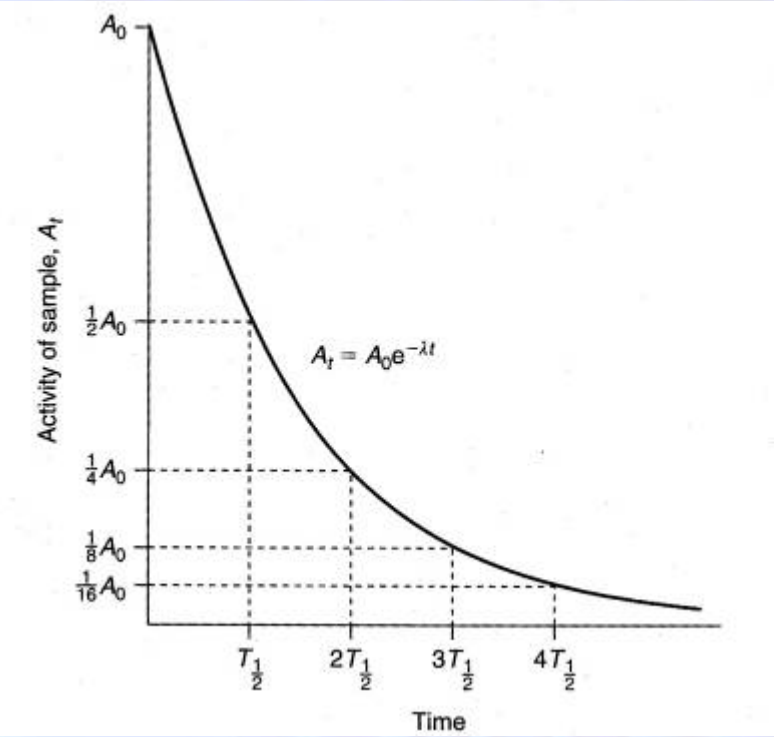
Variation of Activity with Time

- In one half-life activity decays to ½ A_0
- In two half-lives activity decays to ¼ A_0
- Half-life of a particular radioactive isotope is constant and its measurement helps to identify unknown samples



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The Unit of Radioactivity

- Original unit of radioactivity was the curie (Ci)
- The curie was originally related to the activity of 1 gram of radium but it was later standardized to

  $1Ci = 3.7 \times 10^{10}$ disintegrations/sec

- Modern, SI unit is becquerel (Bq)

  1Bq = 1 disintegration/sec

  $1kBq = 10^3$ Bq = 10^3 dis/s

  $1MBq = 10^6$ Bq = 10^6 dis/s

  $1TBq = 10^{12}$ Bq = 10^{12} dis/s

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Interaction of Radiation with Matter

- Charged particles (α, β) lose energy mainly by interacting with atomic electrons. Transferred energy causes either excitation or ionization
- Bremsstrahlung X-rays: released when charged particles slow down rapidly in vicinity of nucleus
- X and γ radiation interactions: photoelectric effect, Compton scattering and pair-production
- Neutrons cannot cause ionization directly. They lose energy through elastic and inelastic scattering and various capture processes

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Penetrating Powers of Nuclear Radiations

- α particles – massive, travel slowly, high probability of interacting with atoms along their path. Lose energy rapidly and only travel very short distances
- β particles - very much smaller than α particles, travel much faster. Undergo fewer interactions per unit length of path and travel further than α's in dense media
- X and γ radiation loses energy mainly by interacting with atomic electrons. Travels very large distances and is very difficult to absorb completely
- Neutrons – interactions are energy dependent. Very penetrating - travel large distances even in dense media

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Radiation Units

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Absorption of Energy

- Energy from a heat source can be absorbed by matter and increase its temperature
- Nuclear radiation can transfer energy from a radiation source to an absorbing medium
- The body can detect harmful levels of heat, but it can not detect absorbed energy from nuclear radiation – even in lethal quantities
- Nuclear radiation differs from heat and other types of radiation in that it has sufficiently high energy to cause ionisation

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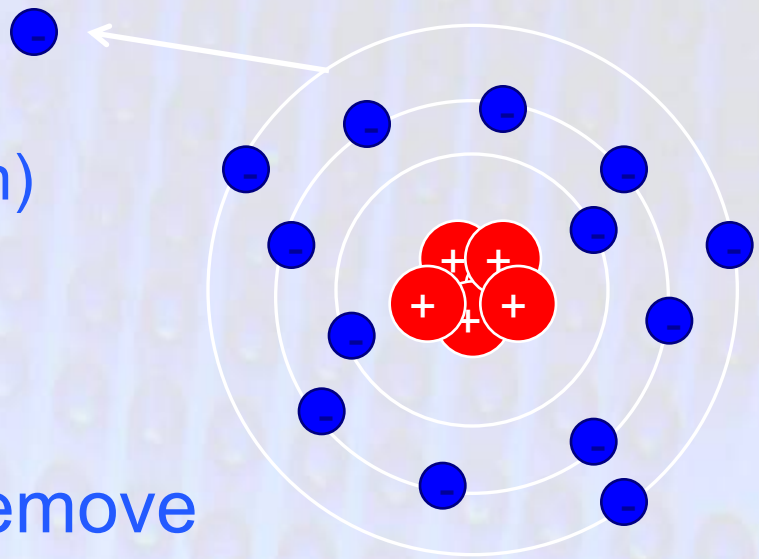
Ionisation

- Removal of an orbital electron from an atom gives
  - an electron
  - remainder of atom (an ion) positively charged
- This is an ion pair
- The energy needed to remove the electron is the ionization energy



Ionisation

- Ionisation energy is supplied by the absorption of radiation energy in the medium
- The radiation loses its energy to the medium in the process
- Alpha, beta, gamma and X-ray radiation is termed ionising radiation
- Ionization in a gas provides a means of detecting radiation

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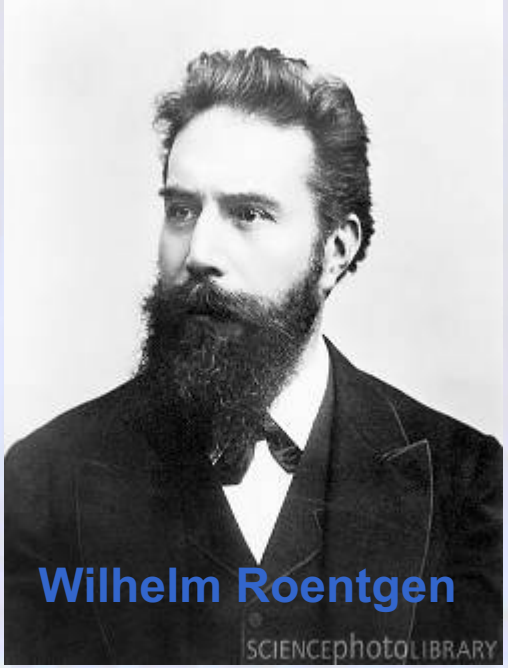
Exposure

The quantity of electronic charge in coulombs (C) produced by ionization per kilogram (kg) of AIR

(either the positive or negative charge – not both)

SI units are C / kg



$$1 \text{ Roentgen} = 2.58 \times 10^{-4} \text{ C / kg}$$

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KERMA

- Kinetic Energy Released per unit MAss
- Units are: Joules per kilogram (J kg^{-1})
- Energy <u>deposited</u> (NOT absorbed) in unit mass of a material (e.g. air) by exposure to radiation
- Only different to Absorbed Dose at high keVs (more than 200 keV) due to:
  - Long range of secondary electrons
  - Bremsstrahlung
- Air KERMA is replacing exposure as standard

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Absorbed Dose (D)

$$\frac{\text{Energy imparted to matter in a small volume (J)}}{\text{Mass of the small volume (kg)}}$$

Hal Gray
Courtesy of the LH Gray Trust

- SI unit is the gray (Gy)
- 1 Gy = 1 Joule of energy absorbed in 1 kg of matter = 1 J/kg

Conversion factor: 1 gray ≈ 100 rads

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Organ or Tissue Dose

$$D_T = \frac{\text{Energy imparted to organ or tissue}}{\text{Mass of the organ or tissue}}$$

More useful for radiation protection purposes

Units: Gray (Gy)

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Converting Dose in Air to Dose in Any Other Tissue

$$\frac{\text{Dose in Tissue}}{\text{Dose in Air}} = \text{Ratio of mass absorption coefficients}$$

Values for mass absorption coefficients can be found in reference books

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Equivalent Dose (H)

$$H = \text{Absorbed Dose (in Grays)} \times \text{Radiation Weighting Factor } (w_R)$$

Strictly speaking this is the absorbed dose averaged over the organ or tissue

Dimensionless quantity

$$\text{Total } H_T = \sum w_R \times D_{T,R}$$

where $D_{T,R}$ is the average absorbed dose to the organ for a particular radiation type

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Equivalent Dose (H)

Unit:

Sievert (Sv)

Still dimensionally J / kg as w_R is just a number

Rolf Sievert – 1929

(from Karolinska Institutet – http://ki.se)

Conversion factor: 1 Sv ≈ 100 rem

Effective Dose (E)

- Accounts for uneven irradiation of the body and represents overall risk from whole body exposure

$$E = \sum_T w_T \times H_T$$

H_T = Equivalent dose to tissue or organ 'T'
w_T = tissue weighting factor

- Tissue weighting factors represent risks of detrimental radiation effects to different organs or tissue

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- Absorbed dose (D): energy absorbed in a medium by any type of ionizing radiation. Unit: Gray 1 Gy = 1 J/kg
- Equivalent dose (H): obtained by multiplying the 'D' by the radiation weighting factor for the particular type of radiation. Unit: Sievert (Sv)
- Radiation weighting factor, w_R: measure of the ability of a particular type of radiation to cause biological damage $w_R = 1$ for β, X and γ, 5 for protons and 20 for α particles
- Effective dose (E): obtained by multiplying the 'H' to each exposed organ by its tissue weighting factor and then summing over all of the organs
- Tissue weighting factor, w_T: reflects the radio-sensitivity of a particular tissue or organ

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- Dose = dose rate × time
- Flux: from point source = $Q/4\pi r^2$
- Committed effective dose: effective dose for internal irradiation
- Annual Limit of Intake (ALI): amount of radionuclide (in Bq) which when taken into the body will result in 20 mSv committed effective dose
- Collective Dose: dose to a particular cohort of persons. Unit: man Sv

The System of Radiological Protection

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Introduction

- ICRP
- Radiation protection philosophy
  - Justification
  - Optimisation
  - Limitation
- Dose limits
- Types of exposure situation
  - Planned exposures
  - Emergency exposures
  - Existing exposures

The role of the ICRP

- The International Commission on Radiological Protection is an internationally recognised independent body, set up in 1928
- Philosophy and standards have evolved
- The 1991 recommendations (Publication 60) described a System of Radiological Protection
- Revised in 2007 (Publication 103) with various mainly minor changes

A reminder of units and quantities (see Chapter 3)

Absorbed dose D (Gy)

Equivalent dose H (Sv) in tissue T where

$$H_T = D_T \times w_R$$

w_R is the radiation weighting factor

Effective dose E (Sv) where

$$E = \sum_T w_T \sum_R w_R D_{T,R}$$

w_T is the tissue weighting factor

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Radiation weighting factors

Type of radiation	wR
X-rays, γ-rays and electrons	1
Protons	5
Thermal neutrons	2.5
Fast neutrons	2.5 to 20 *
Alpha particles, fission fragments	20

* Depending on energy

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Tissue weighting factors

Organ	w_T for organ
Gonads	0.08
Red bone marrow, colon, lung, stomach, breast	0.12
Bladder, liver, oesophagus, thyroid	0.04
Skin, bone surface, brain, salivary glands	0.01
Remainder (in total)	0.12

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Dose Limits

- ICRP recommendations used the term tolerance dose – but this seemed to imply a threshold level below which no effects
- Became maximum permissible dose
- Now use term dose limit
- Emphasis on is on ensuring all doses as low as reasonably achievable (ALARA)

The 2007 Recommendations
(ICRP 103)

Recognise three exposure situations

- Planned – practice deliberately undertaken (e.g. nuclear facilities, medical exposure)
- Emergency – where urgent action needed to avoid or reduce exposure (e.g. accidents and incidents)
- Existing – where source is pre-existing (e.g. natural sources or after-effects of accident)

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Planned exposure situations

The system of protection is based on

- Justification – benefit should offset detriment: i.e. do more good than harm
- Optimisation – numbers of people exposed and levels of exposure kept **ALARA**
- Dose and risk limits for individuals

(Note that no risk limits are specified but implication is that they should be commensurate with those corresponding to dose limits)

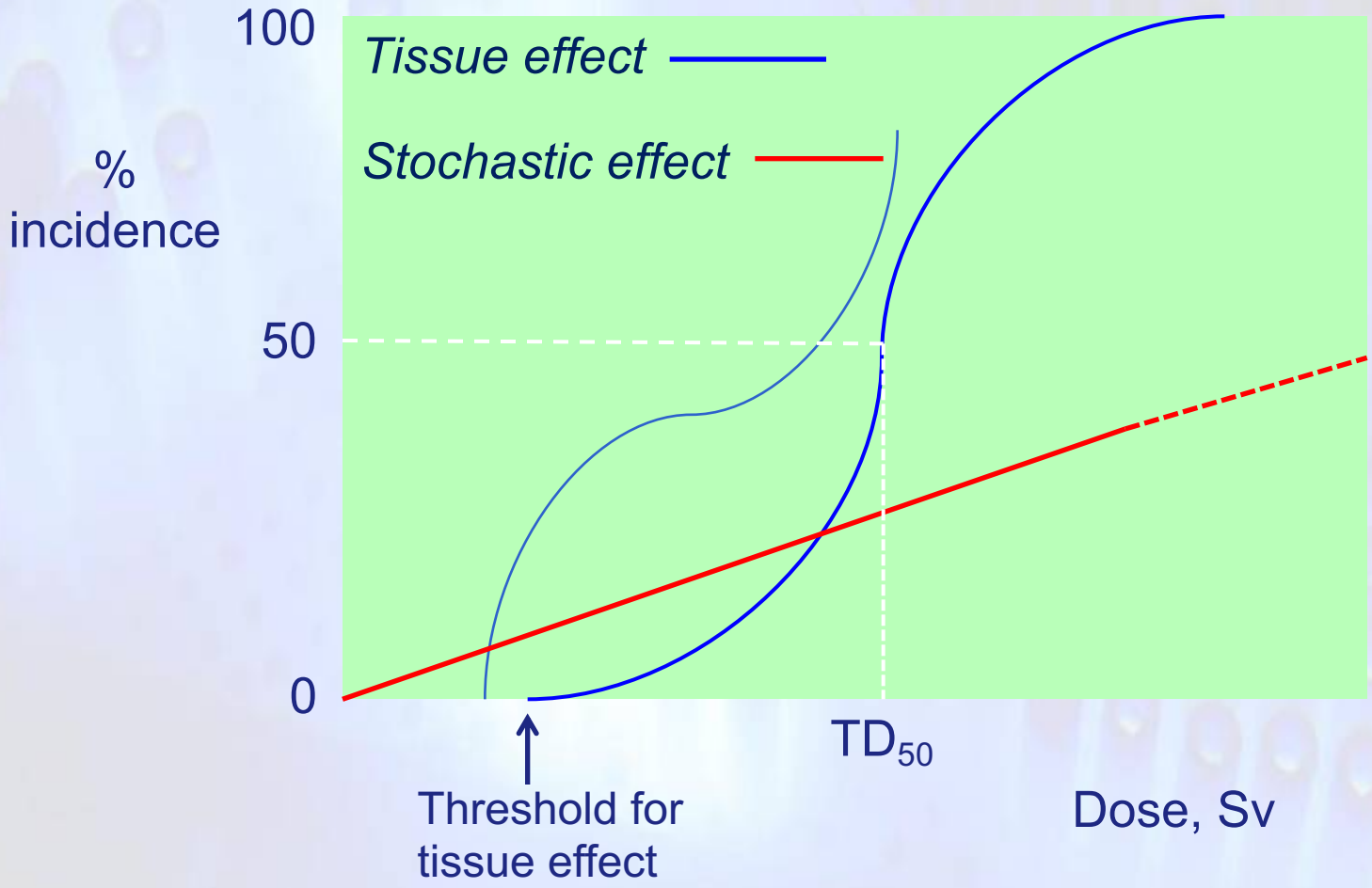
Basis of dose limits

- In Publication 103 the ICRP reviews and gives quantitative estimates of the consequences of radiation exposure. The effects are of two types:
  - Harmful tissue reactions: severity of effect increases with dose – usually only occur above threshold (e.g. acute effects)
  - Stochastic effects: probability of effect increases with dose (without threshold) e.g. induction of cancer

Basis of dose limits

- The aim is to prevent harmful tissue reactions and to limit the probability of stochastic effects to acceptable levels
- This is achieved by:
  - setting dose limits so that no threshold dose is reached (prevention of harmful tissue reactions)
  - keeping all exposures ALARA (limitation of stochastic effects)

Illustration of dose response



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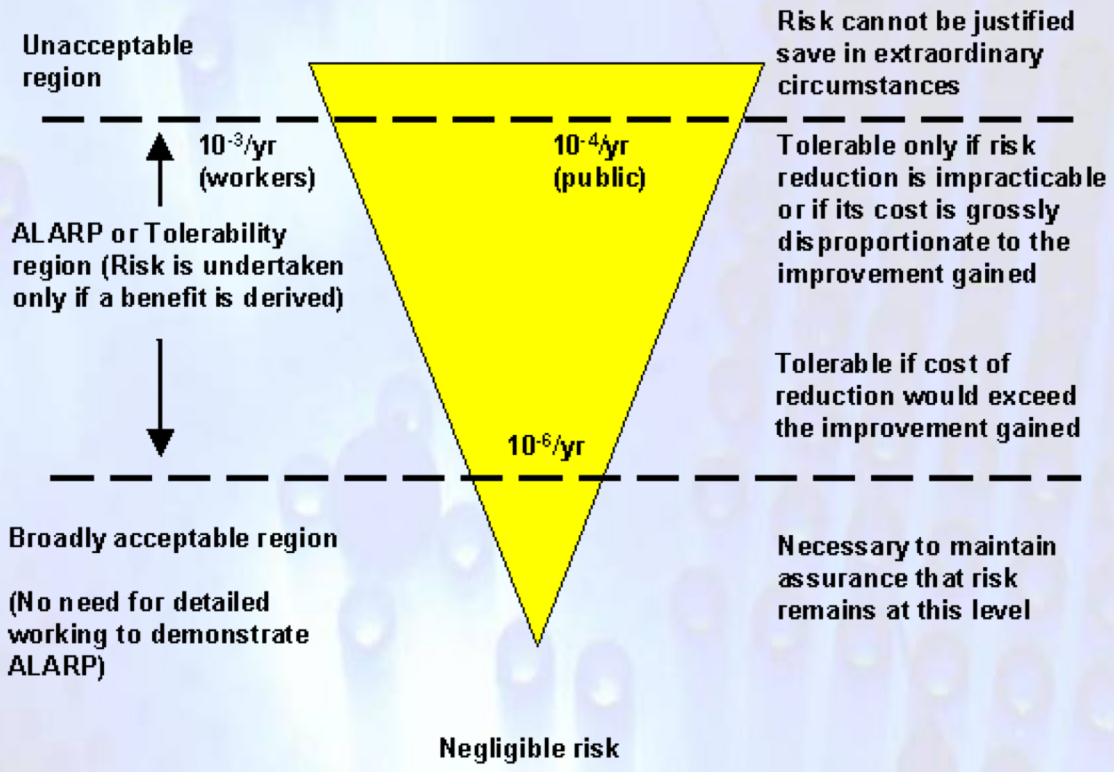
Tolerability and acceptability

- ICRP recognises three levels of tolerability of exposure or risk:
  - Unacceptable: level not acceptable in planned situations
  - Tolerable: means that the exposure can be tolerated where there is sufficient benefit
  - Acceptable: the level of protection has been optimised

Tolerability and acceptability



(Health and Safety Executive, UK)

Dose limits and acceptability

- The dose limits recommended by the ICRP are set in the region between the tolerable and the unacceptable levels
- Therefore, at dose limits, continued exposure is only just tolerable

Dose limits for workers

- To limit stochastic effects, the annual effective dose limit for uniform irradiation of the whole body is 20 mSv
- This may be averaged over 5 years but the dose in any one year should not exceed 50 mSv

Recommended dose limits

	Dose limit	
	Occupational	**Public**
Effective dose	20 mSv per year	1 mSv in a year
Annual equivalent dose in:		
Lens of eye	20 mSv	15 mSv
Skin	500 mSv	50 mSv
Hands and feet	500 mSv	-

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Notes on dose limits for workers

- All unnecessary exposures should be avoided
- The average dose to a group of workers is normally only a few mSv. At this level, the average risk is comparable with that in ‘safe’ industries
- Same for women but once a pregnancy is declared, the foetus should receive same level of protection as member of public

Workplace classification

Work places subject to classification:

Controlled area – in which normal working conditions require workers to follow well-established procedures

Supervised area – where no special procedures are normally needed but exposure conditions are kept under review

Dose limits for public

- The dose limit is 1 mSv for members of the public
- In special circumstances a higher value could be allowed provided average over 5 years does not exceed 1 mSv
- For prevention of tissue effects, ICRP recommends limits of 15 mSv/y for lens of eye and 50 mSv/y for skin

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Planned exposure

- This is the regime within which radiation protection is normally carried out, e.g. in nuclear facilities or medical and teaching establishments
- Based on justification, optimisation and dose limitation

Justification of practices

- Need to show that any new practice will have a positive net benefit
- For example, the case for nuclear power needs to show that benefits are sufficient to justify the additional radiation exposure and risk to workers and general population
- Similarly for any new medical diagnostic or treatment technique

Justification of practices

- Justification is often of a generic nature and undertaken at national level
- Also situations when it needs to be considered at local level, particularly for medical exposures. Often needs to be considered at level of individual patient

Optimisation

- Optimisation requires that exposures are reduced as far below dose limits as reasonably achievable
- ICRP recommends use of the concept of dose and risk constraints applied to exposure of individual from a single source
- Where there are several sources of exposure, constraints on each source would ensure dose limit not exceeded

Dose limitation

- Important to appreciate that, in planned situations, the dose limits are absolute upper limits on exposure
- Application of ALARA usually results in doses well below limits
- Experience shows that for workers the average dose is generally 10% or less of the limits

Emergency exposure

- Normal dose limits do not apply
- Actions based on justification and optimisation
- Very important to have a detailed and well-rehearsed emergency plan
- Assisted by the use of pre-defined intervention and reference levels though these only provide useful guidelines

Emergency exposure - workers

- Informed volunteers may receive large doses for purpose of saving life, preventing serious injury or preventing major escalation in scale of incident
- For saving of life, doses of up to about 1 Sv could be justifiable
- At Fukushima, for workers seeking to control the situation, a reference level of 100 mSv was set initially but this was later raised to 250 mSv

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Emergency exposure - public

- Need framework of intervention levels to assist decisions on application of counter-measures aimed at reducing exposures
- These can include:
  - sheltering indoors
  - evacuation
  - administration of stable iodine
  - control of food and water
- Decisions to apply countermeasures should take account of the associated detriments

Existing exposure situations

Examples are:

- High level of natural radiation in the environment or within buildings (e.g. radon) as a result of local geology
- High levels of radioactivity on ground due to previous nuclear accident e.g. Cs-137 contamination of agricultural land

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Existing exposure situations

- Can involve large areas and large populations
- Decisions on countermeasures should take account of practical considerations and of their disruptive effects
- The ICRP approach is to encourage national authorities to establish reference levels to assist decision-making whilst also allowing flexibility